

Probability Distribution

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Random Variable

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A random variable is a variable that takes on **numerical values** as a result of a random experiment or measurement; associates a numerical value with each possible outcome.

The differences between variable and random variable are-

- Random variable always takes **numerical values**
- There is a **probability associated** with each possible values

Random variable is denoted by capital letters such as X, Y, Z etc.

And the possible outcomes are denoted by small letters such as x, y, z etc.

Random Variable

Example 1:

A coin is tossed. It has two possible outcomes- Head and Tail.

Consider a variable, $X = \text{outcome of a coin toss} = \begin{cases} H, & \text{if Head appears} \\ T, & \text{if Tail appears} \end{cases}$

Here, $S = \{H, T\}$.

But, these are not numerical values.

Random Variable

Example 1(contd.):

Consider a variable, X = Number of heads obtained in a trial

$$\text{Then, } X = \begin{cases} 1, & \text{if Head appears} \\ 0, & \text{if Tail appears} \end{cases}$$

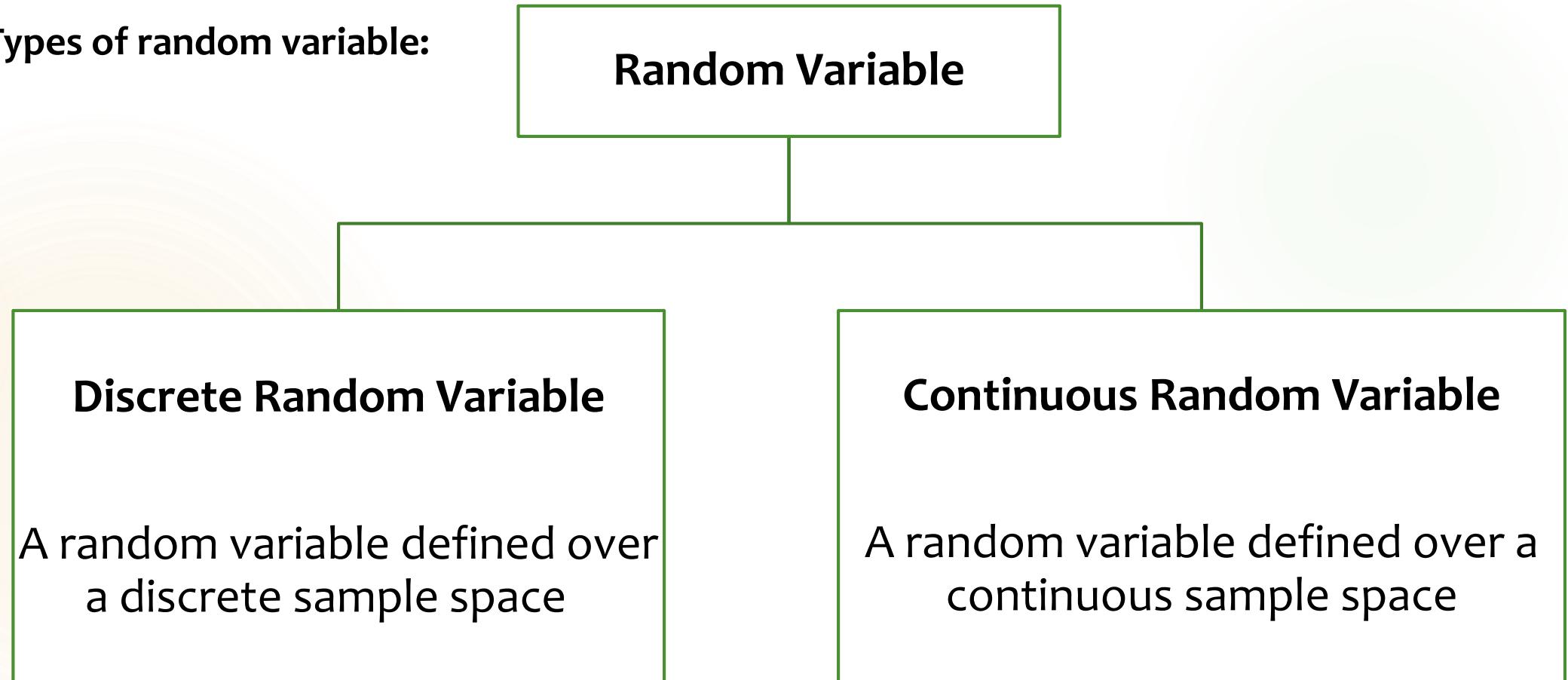
For a fair coin, we can write, $P(X=1) = \frac{1}{2}$ and $P(X=0) = \frac{1}{2}$

So, X is a random variable.

Random Variable

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Types of random variable:



Random Variable

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Examples:

Discrete Random Variable:

1. X = Number of correct answers in a 100-MCQ test = $0, 1, 2, \dots, 100$
2. X = Number of cars passing a toll both in a day = $0, 1, 2, \dots, \infty$
3. X = Number of balls required to take the first wicket = $1, 2, 3, \dots, \infty$

Continuous Random Variable:

1. X = Weight of a person. $0 < X < \infty$
2. X = Monthly Profit. $-\infty < X < \infty$

Probability Distributions

Distribution of the probabilities among the different values of a random variable.

Discrete probability distribution- probability distribution of a discrete random variable

Continuous probability distribution- probability distribution of a continuous random variable

Probability Distributions

Examples:

Discrete probability distribution-

- Tossing a coin 2 times.

X = Number of Heads appeared

$S = \{HH, HT, TH, TT\}$

x	0	1	2
$P(x)$	$\frac{1}{4}$	$\frac{2}{4}$	$\frac{1}{4}$

Probability Distributions

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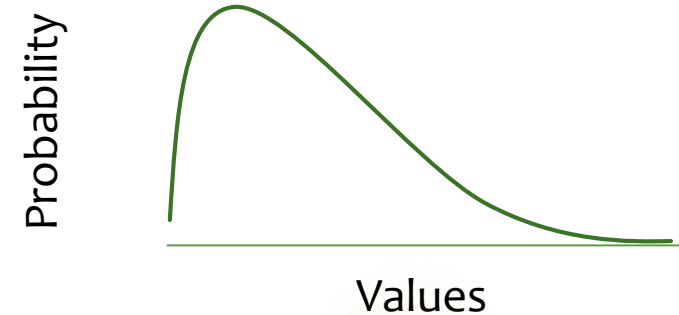
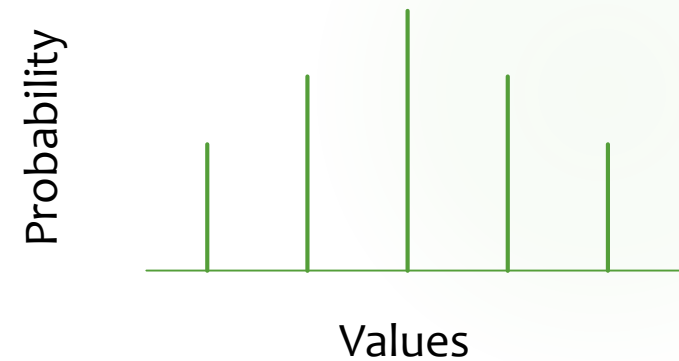
Different types of probability distributions:

Discrete probability distribution-

1. Bernoulli Distribution
2. *Binomial Distribution*
3. Poisson Distribution etc.

Continuous probability distribution-

1. Uniform Distribution
2. *Normal Distribution*
3. Exponential Distribution
4. t-distribution etc.



PMF and PDF

Probability Mass Function (pmf)- the probability distribution function of a discrete random variable X is called a pmf and is denoted by $p(x)$

Probability Density Function (pdf)- the probability distribution function of a continuous random variable X is called a pdf and is denoted by $f(x)$

Binomial Distribution

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Bernoulli trial:

A trial that has only two possible outcomes (often called 'Success' and 'Failure')

Outcome	Success	failure
Probability	p	$1-p$

Let,

- n independent Bernoulli trials are performed
- Each trial has the same probability of success, p

Binomial Distribution

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Let,

X= number of success in n trials

Then, X is a binomial random variable with distribution function (pmf),

$$p(x) = {}^nC_x p^x (1-p)^{n-x} \quad ; x = 0, 1, 2, \dots, n$$
$$= \frac{n!}{(n-x)! x!} p^x (1-p)^{n-x}$$

Here, $n! = n(n-1)(n-2) \dots 1$ $0! = 1$ $1! = 1$ $2! = 2 \times 1 = 2$ $3! = 3 \times 2 \times 1 = 6$

We write it as, $X \sim \text{binomial}(n, p)$

Binomial Distribution

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Mean of the binomial distribution, $\mu = E(X) = \sum x p(x) = np$

Variance of binomial distribution, $\sigma^2 = E(X^2) - \mu^2 = np(1 - p) = npq$

Standard deviation of binomial distribution, $\sigma = \sqrt{npq}$

Binomial Distribution

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Example 3:

There are 3 multiple choice questions in a MCQ test. Each MCQ consists of four possible choices and only one of them is correct. If an examinee answers those MCQ randomly (without knowing the correct answers)

- a. What is the probability that exactly any two of the answers will be correct?
- b. What is the probability that at least two of the answers will be correct?
- c. What is the probability that at most two of the answers will be correct?
- d. What will be the average or expected number of correct answers?
- e. Also, find the standard deviation of number of correct answers.

Binomial Distribution

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Solution:

Let,

X= number of correct answers selected in 3 MCQs

Here, p = probability of selecting correct answer per question = $\frac{1}{4} = 0.25$

$$\therefore X \sim \text{binomial } (n = 2, p = 0.25)$$

$$\begin{aligned} p(x) &= {}^3C_x (0.25)^x (1 - 0.25)^{3-x} && ; x = 0, 1, 2, 3 \\ &= \frac{3!}{(3-x)! x!} (0.25)^x (0.75)^{3-x} \end{aligned}$$

Binomial Distribution

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Solution (contd.):

a. probability that exactly any two of the answers will be correct-

$$\begin{aligned} P(X = 2) &= \frac{3!}{(3-2)! 2!} (0.25)^2 (0.75)^{3-2} \\ &= \frac{3!}{1! 2!} (0.25)^2 (0.75)^1 = \frac{3 * 2 * 1}{1 * (2 * 1)} * 0.0625 * 0.75 = 0.141 \end{aligned}$$

b. probability that at least two of the answers will be correct-

$$\begin{aligned} P(X \geq 2) &= P(X = 2) + P(X = 3) \\ &= \frac{3!}{(3-2)! 2!} (0.25)^2 (0.75)^{3-2} + \frac{3!}{(3-3)! 3!} (0.25)^3 (0.75)^{3-3} \\ &= \frac{3!}{1! 2!} (0.25)^2 (0.75)^1 + \frac{3!}{0! 3!} (0.25)^3 (0.75)^0 = 0.141 + 0.016 = 0.157 \end{aligned}$$

Binomial Distribution

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Solution (contd.):

c. probability that at most two of the answers will be correct-

$$\begin{aligned} P(X \leq 2) &= P(X = 0) + P(X = 1) + P(X = 2) \\ &= \frac{3!}{(3-0)!0!} (0.25)^0 (0.75)^{3-0} + \frac{3!}{(3-1)!1!} (0.25)^1 (0.75)^{3-1} \\ &\quad + \frac{3!}{(3-2)!2!} (0.25)^2 (0.75)^{3-2} \\ &= \frac{3!}{3!0!} (0.25)^0 (0.75)^3 + \frac{3!}{2!1!} (0.25)^1 (0.75)^2 + \frac{3!}{1!2!} (0.25)^2 (0.75)^1 \\ &= 0.422 + 0.422 + 0.141 = 0.985 \end{aligned}$$

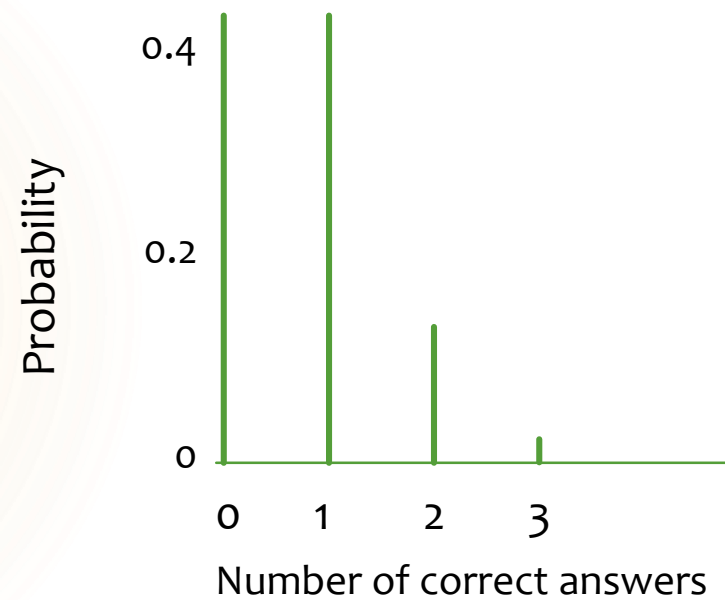
d. $E(X) = np = 3 * .25 = 0.75$

e. $SD(X) = \sqrt{npq} = \sqrt{3 * 0.25 * 0.75} = 0.75$

Binomial Distribution

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X	0	1	2	3
P(x)	0.422	0.422	0.141	0.016



Poisson Distribution

If an event occur within a time interval/place
example: no.of accident

Let,

X = a random variable usually counts or number of occurrences

Then, X is a Poisson random variable with distribution function (pmf),

$$p(x) = \frac{e^{-\lambda} \lambda^x}{x!} \quad ; x = 0, 1, 2, \dots$$

We write it as, $X \sim \text{Poisson}(\lambda)$

Poisson Distribution

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Mean of the Poisson distribution, $\mu = E(X) = \sum x p(x) = \lambda$

Variance of Poisson distribution, $\sigma^2 = E(X^2) - \mu^2 = \lambda$

Standard deviation of Poisson distribution, $\sigma = \sqrt{\lambda}$

Example

$$P(x, \text{lamda}) = P(2, 2.5) =$$

Example: Telephone calls arrive at a switchboard at a mean rate of 0.5 calls per minute. Calculate the probability that two calls will arrive in a particular five minute period.

Solution: $\lambda = np = 5 \times 0.5 = 2.5$

$$P(2, 2.5) = \frac{e^{-2.5} (2.5)^2}{2!} = 0.257$$

Poisson Distribution

Example 4:

The average number of errors on a page of a certain magazine is 0.2.
What is the probability that the next page (or a randomly selected page) you read contains

- i. 0 (zero) error?
- ii. 2 or more errors?
- iii. What is the average error per page?
- iv. Also, find standard deviation of the number of errors.

Poisson Distribution

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Solution:

Let,

X= number of errors in a page

Here, λ = average number of errors per page= 0.2

$$\therefore X \sim \text{Poisson} (\lambda = 0.2)$$

$$\begin{aligned} p(x) &= \frac{e^{-\lambda} \lambda^x}{x!} \quad ; x = 0, 1, 2, \dots \\ &= \frac{e^{-0.2} 0.2^x}{x!} \end{aligned}$$

Poisson Distribution

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Solution:

i.
$$P(X = 0) = \frac{e^{-0.2} 0.2^x}{x!} = \frac{e^{-0.2} 0.2^0}{0!} = \frac{e^{-0.2} * 1}{1} = 0.8187$$

ii.
$$\begin{aligned} P(X \geq 2) &= 1 - P(X < 2) = 1 - [P(X = 0) + P(X = 1)] \\ &= \frac{e^{-0.2} 0.2^0}{0!} + \frac{e^{-0.2} 0.2^1}{1!} = 1 - [e^{-0.2} + e^{-0.2} * 0.2] = 0.01756 \end{aligned}$$

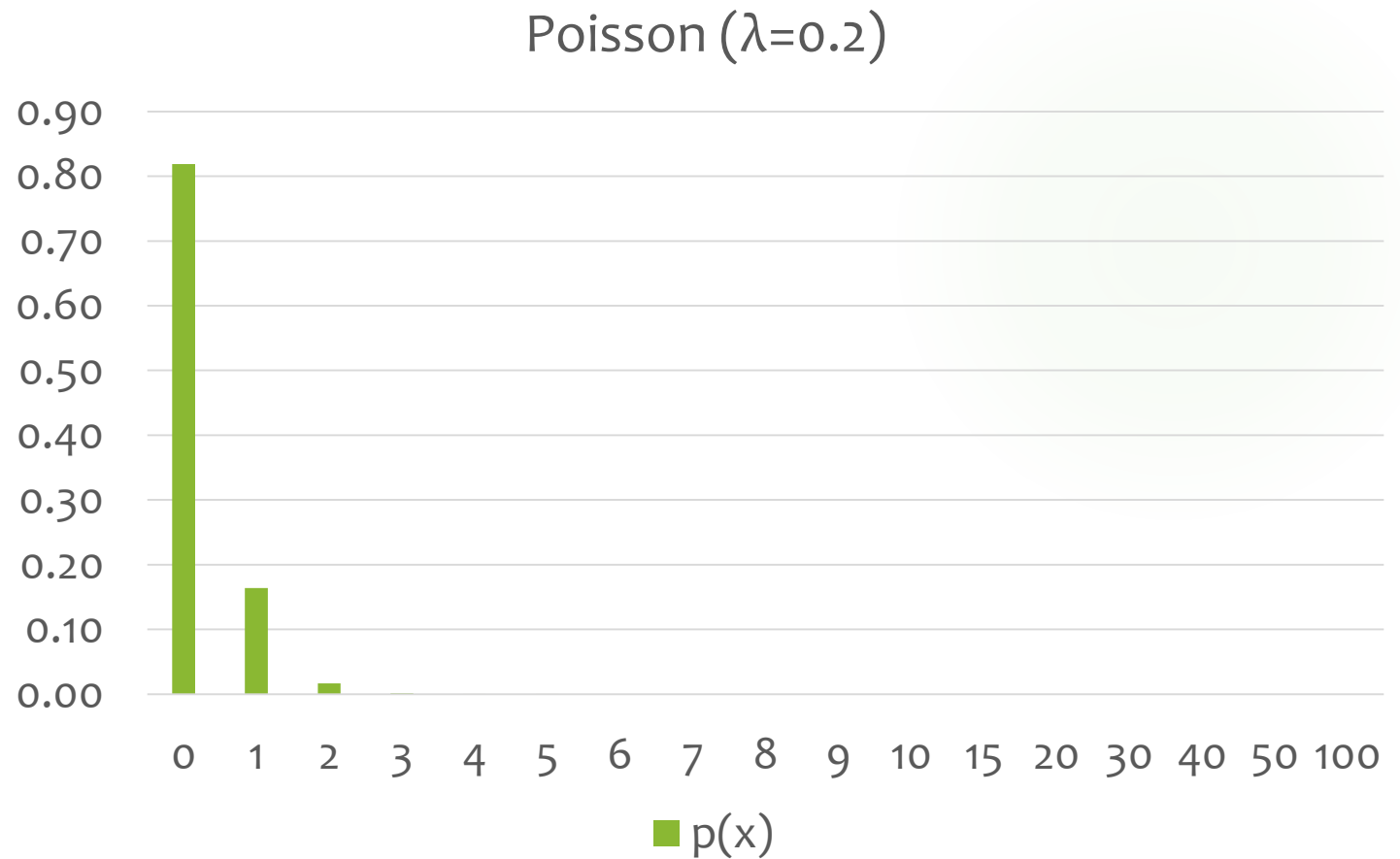
iii. Average number of errors, $E(X) = \lambda = 0.2$

iv. Standard deviation, $SD(X) = \sqrt{\lambda} = \sqrt{0.2} = 0.45$

Poisson Distribution

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x	p(x)
0	0.82
1	0.16
2	0.02
3	0.00
4	0.00
5	0.00
6	0.00
7	0.00
8	0.00
9	0.00
10	0.00
15	0.00
20	0.00
30	0.00
40	0.00
50	0.00
100	0.00



Uniform Distribution

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Let,

X is a continuous random variable

Then, if X has a probability density function (pdf),

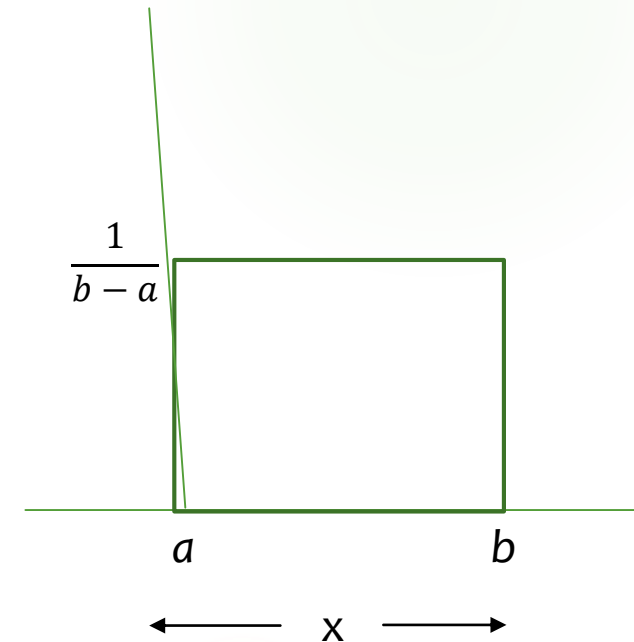
$$f(x) = \frac{1}{b-a} \quad ; \quad a < x < b$$

$$A=1, b=10 \quad f(x)=1/9 \quad f(x=2)=1/9 \quad f(3.1)=1/9$$

We write it as, $X \sim \text{uniform}(a, b)$

$$\text{Mean, } E(X) = \frac{a+b}{2}$$

$$\text{Variance, } V(X) = \frac{(b-a)^2}{12}$$



Uniform Distribution

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Example 5:

The waiting time (in minutes) for train is uniform (10, 50).

Find-

- a. The probability that you have to wait at least 20 minutes. $P(x \geq 20)$
- b. Average waiting time.
- c. Standard deviation of waiting time.

Uniform Distribution

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Solution:

Let,

X= waiting time (in minutes)

$$\therefore X \sim \text{uniform}(10, 50)$$

$$\begin{aligned} f(x) &= \frac{1}{b-a} ; a < x < b \\ &= \frac{1}{50-10} ; 10 < x < 50 \\ &= \frac{1}{40} \end{aligned}$$

Uniform Distribution

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Solution:

- a. Probability that you have to wait at least 20 minutes-

$$\begin{aligned} P[X \geq 20] &= \int_{20}^{50} f(x) dx = \int_{20}^{50} \frac{1}{40} dx = \frac{1}{40} \int_{20}^{50} 1 dx \\ &= \frac{1}{40} [x]_{20}^{50} = \frac{1}{40} [50 - 20] = \frac{30}{40} = \frac{3}{4} = 0.75 \end{aligned}$$

- b. Average waiting time-

$$E(X) = \frac{a + b}{2} = \frac{10 + 50}{2} = \frac{60}{2} = 30 \text{ minutes}$$

Uniform Distribution

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Solution:

c. Standard deviation of waiting time-

$$SD(X) = \sqrt{\frac{(b - a)^2}{12}} = \sqrt{\frac{(50 - 10)^2}{12}} = \sqrt{\frac{(40)^2}{12}} = 3.65 \text{ minutes}$$

Normal Distribution

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Let,

X is a continuous random variable

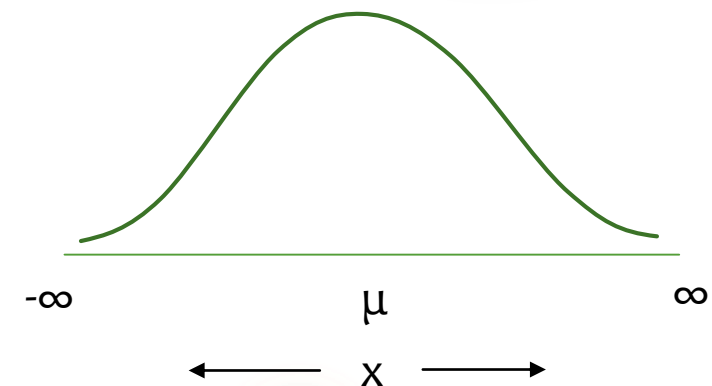
Then, if X has a probability density function (pdf),

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2} ; -\infty < x < \infty$$

We write it as, $X \sim N(\mu, \sigma^2)$

Mean, $E(X) = \mu$

Variance, $V(X) = \sigma^2$



Standard Normal Distribution

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Let,

$$Z = \frac{X - \mu}{\sigma}$$

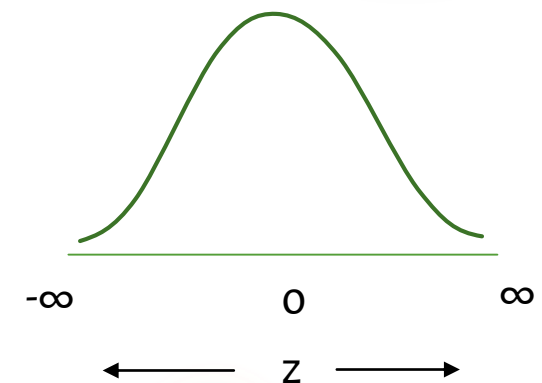
Then, Mean, $E(Z) = 0$

Variance, $V(Z) = 1$

And, if Z has a probability density function (pdf),

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} ; -\infty < z < \infty$$

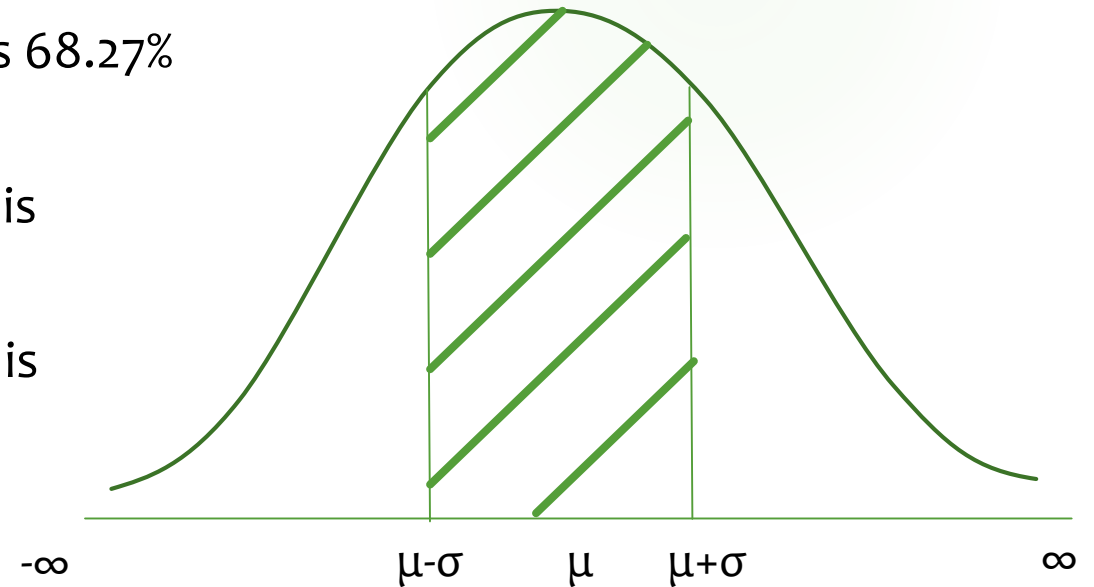
We write it as, $Z \sim N(0, 1)$



Characteristics of a Normal Distribution

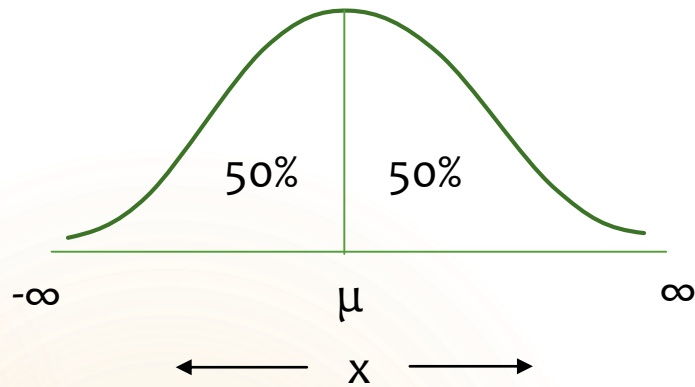
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1. Mean= Median = Mode
2. Symmetric and Mesokurtic
3. Bell-shaped curve
4. The area under the curve lying between $\mu \pm \sigma$ is 68.27% of the total area
5. The area under the curve lying between $\mu \pm 2\sigma$ is 95.45% of the total area
6. The area under the curve lying between $\mu \pm 3\sigma$ is 99.73% of the total area



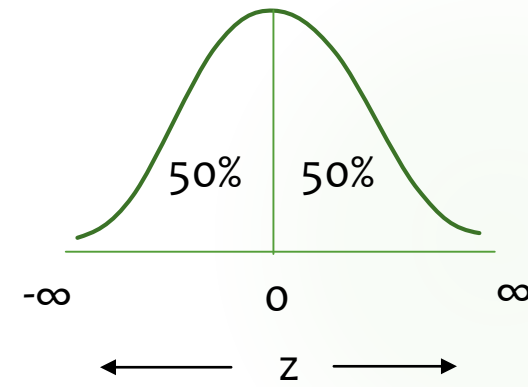
Characteristics of a Normal Distribution

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$$P[X < \mu] = P[X > \mu] = 0.5$$

$$P[X < -x] = P[X > x]$$



$$P[Z < 0] = P[Z > 0] = 0.5$$

$$P[Z < -z] = P[Z > z]$$

Normal Distribution Table

Z-table

- Normal distribution table provides probabilities for $N(0,1)$ i.e. for standard normal distribution
- Usually, normal table gives $P[0 < Z < z]$ for positive values of Z .
- For other values, we can use the property of symmetry with median 0 of standard normal distribution
- To find probabilities for a normal random variable X , we can transform the probability statement about X in terms of probability statement for Z and then calculate the probability using the standard normal distribution table or Z-table

$$P[X < a] = P\left[\frac{X - \mu}{\sigma} < \frac{a - \mu}{\sigma}\right] = P\left[Z < \frac{a - \mu}{\sigma}\right]$$

Finding Area Under the Normal Curve using Z-table

Example 6:

The number of viewers of a TV show per week has a mean of 29 million with a standard deviation of 5 million. Assume that, the number of viewers of that show follows a normal distribution.

What is the probability that, next week's show will-

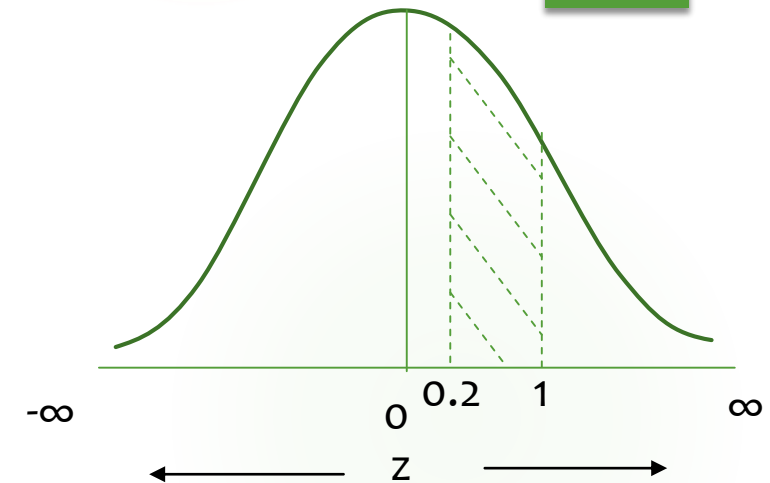
- a. Have between 30 and 34 million viewers?
- b. Have at least 23 million viewers?
- c. Exceed 40 million viewers?

Finding Area Under the Normal Curve using Z-table

Solution:

Let, X = Number of viewers of the show per week (in million)

$$\therefore X \sim N(\mu, \sigma^2)$$



- a. the probability that, next week's show will have between 30 and 34 million viewers-

$$\begin{aligned} P[30 \leq X \leq 34] &= P\left[\frac{30 - \mu}{\sigma} \leq \frac{X - \mu}{\sigma} \leq \frac{34 - \mu}{\sigma}\right] = P\left[\frac{30 - 29}{5} \leq \frac{X - \mu}{\sigma} \leq \frac{34 - 29}{5}\right] \\ &= P[0.20 \leq Z \leq 1] = P[0 \leq Z \leq 1] - P[0 \leq Z \leq 0.2] = 0.3413 - 0.0793 \\ &= 0.262 \end{aligned}$$

Finding Area Under the Normal Curve using Z-table

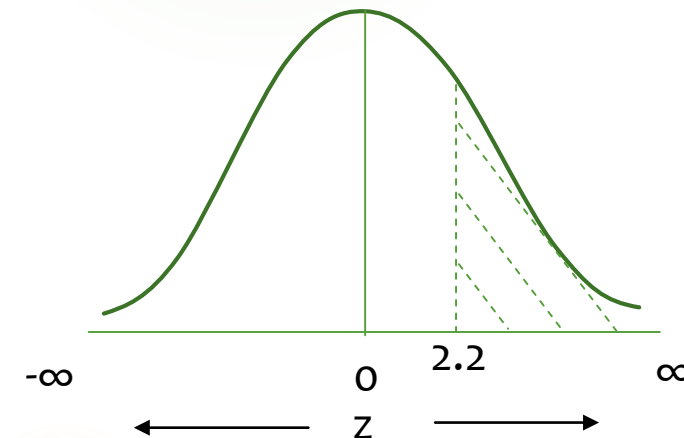
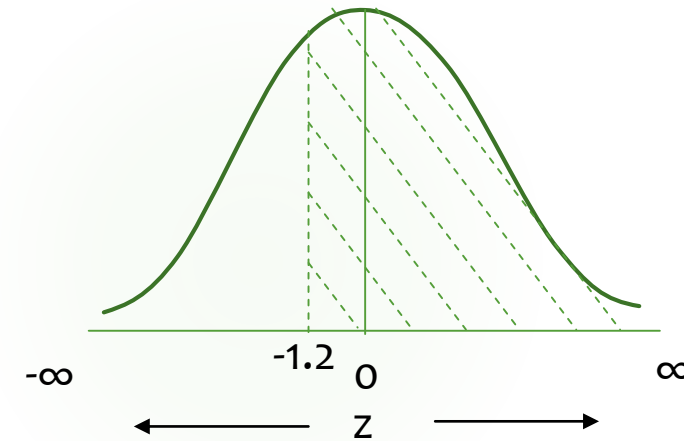
Solution (contd.):

b. the probability that, next week's show will have at least 23 million viewers-

$$\begin{aligned} P[X \geq 23] &= P\left[\frac{X - \mu}{\sigma} \geq \frac{23 - \mu}{\sigma}\right] = P\left[\frac{X - \mu}{\sigma} \geq \frac{23 - 29}{5}\right] \\ &= P[Z \geq -1.2] = P[-1.2 \leq Z \leq 0] + P[Z \geq 0] = 0.3849 + 0.5 \\ &= 0.8849 \end{aligned}$$

c. the probability that, next week's show will exceed 40 million viewers-

$$\begin{aligned} P[X > 40] &= P\left[\frac{X - \mu}{\sigma} > \frac{40 - \mu}{\sigma}\right] = P\left[\frac{X - \mu}{\sigma} > \frac{40 - 29}{5}\right] \\ &= P[Z > 2.2] = P[Z \geq 0] - P[0 \leq Z \leq 2.2] = 0.5 - 0.4861 = 0.0139 \end{aligned}$$



Standard normal probabilities (continued)

[illegible]

z	.00	.01	.02	.03	.04	.05	.06 \	.07	.08	.09
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	<u>.4325</u>	.4286	.4247
0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

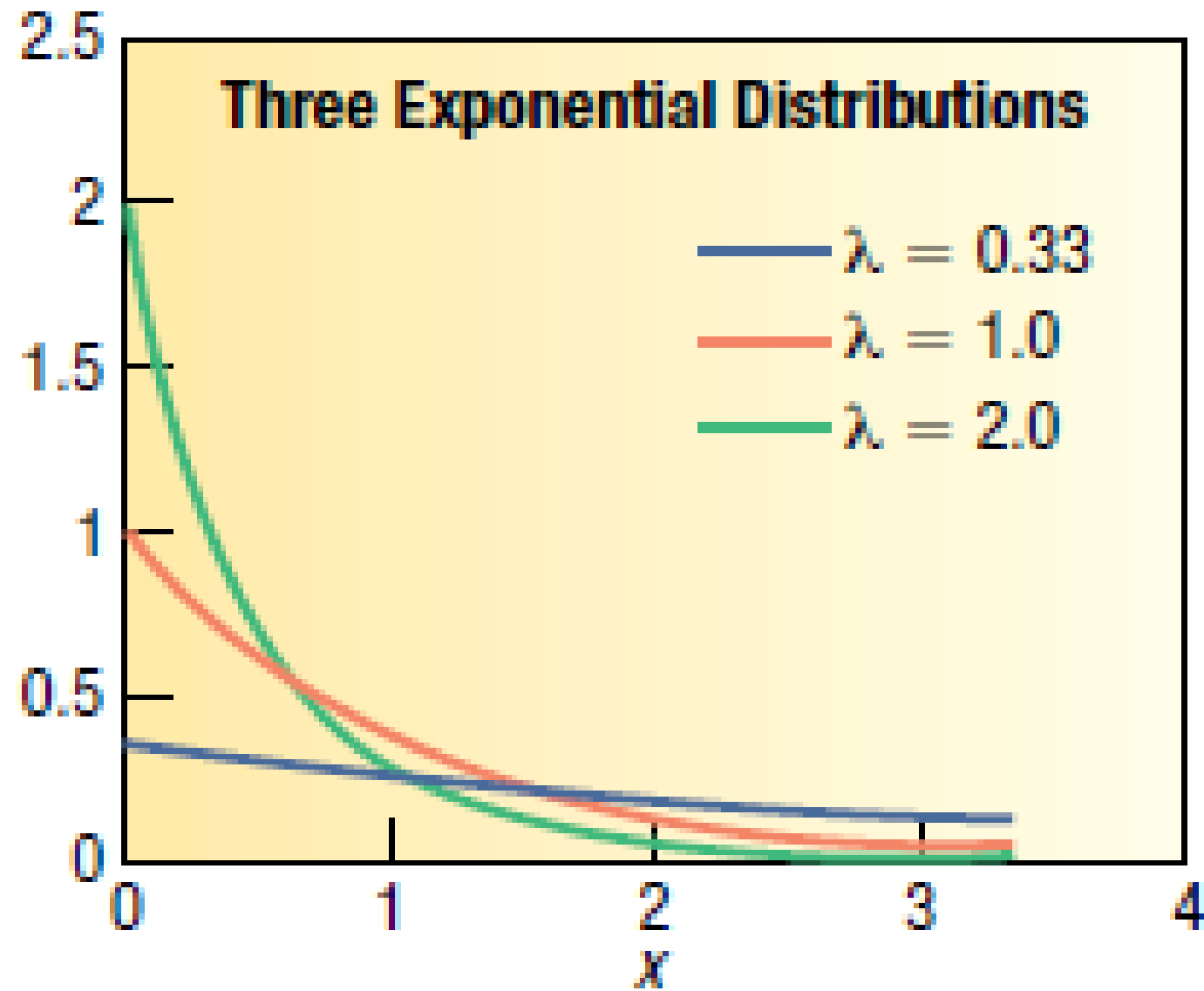
The Family of Exponential Distributions

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- ▶ This continuous probability distribution usually describes times between events in a sequence.
- ▶ The actions occur independently at a constant rate per unit of time or length.

The service times in a system (e.g., how long it takes to serve a customer).

- The time between “hits” on a website.
- The lifetime of an electrical component.
- The time until the next phone call arrives in a customer service center.



- ▶ λ is often referred to as the “rate” parameter
- ▶ The exponential probability distribution is positively skewed

EXPONENTIAL DISTRIBUTION

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$$P(x) = \lambda e^{-\lambda x}$$

Cumulative DISTRIBUTION

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$$P(\text{Arrival time} < x) = 1 - e^{-\lambda x}$$

- ▶ Orders for prescriptions arrive at a pharmacy website according to an exponential probability distribution at a mean of one every 20 seconds.
- ▶ Find the probability the next order arrives in less than 5 seconds,
- ▶ in more than 40 seconds, or
- ▶ Between 5 and 40 seconds.

$$P(\text{Arrival time} < 5) = 1 - e^{-\frac{1}{20}(5)}$$

$$P(5 \leq x \leq 40) = P(\text{Arrival} \leq 40) - P(\text{Arrival} \leq 5)$$

Math

- ▶ The lifetime of plasma and LCD TV sets follows an exponential distribution with a mean of 100,000 hours. Compute the probability a television set:
 - ▶ **a. Fails in less than 10,000 hours.**
 - ▶ **b. Lasts more than 120,000 hours.**
 - ▶ **c. Fails between 60,000 and 100,000 hours of use.**